Chater 4

← Lists and Tuples →

Python Lists are containers to store a set of values of any data type.

friends = ["Apple", "Akash", "Rehan", 7, False]

str() int() bool()

List Indexing - A List can be indexed just lik a string

L1 = [7, 9, "Horse"]

L1[0] => 7
L1[1] => 1

L1[70] => Error
L1[0:2] => [7, 9] → List slicing

List methods - consider the following list:

L1 = [1, 8, 7, 2, 21, 15]

1> L1.Sort() : updates the list to [1, 2, 7, 8, 15, 21]
2> L1.reverse(): updates the list to [15, 21, 2, 7, 8, 1]
3> L1.append(8) adds 8 at the end of the list
4> L1.insert(3,8) : This will add 8 at 3 index.
5> L1.pop(2) : Will delete element at index 2 and return its value.
6> L1.remove(21) : Will remove 21 from the list.

Tuples in Python - A Tuple is an immutable data type in python - cannot change

a = () => Empty tuple
a = (1,) => Tuple with only one element needs a c
a = (1,7,2) => Tuple with more than one element

Once defined a Tuple' element cant be altere or manipulated.

Tuple methods - consider the following tuple

a = (1,7,2)

1> a.count(1): a.count(1) will return number of times 1 occurs in a.

2> a.index(1): a index(1) will return the index of first occurence of 1 in a.

Chapter 4 - Practice Set

1. Write a Program to store seven fruits in a list entered by the user.

2. Write a program to accept marks of 5 students and display them in a sorted manner.

3. Check that a tuple cannot be changed in Python.

4. Write a program to sum a list with 4 numbers.

5. Write a program to count the number of zeros in the following tuple:

a = (7, 0, 8, 0, 0, 9)

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EXPLORER

CHAPTER-4_LISTS_AND_TUPLES

- 1_lists.py
- methods.py
- practiceSet-4.ipynb
- tuple0.py

OUTLINE

TIMELINE

1_lists.py > ...

```python
friends=["virendra","pradeep",12,True,5.6]
print(friends)
print(friends[4])
friends[4]="xyz" # lists are mutable
print(friends)
friends[0:]

# lists methods
list1=[1,2,3,10,9,5,6,7,8]

list1.sort()
list1.reverse()
print(list1)
```

tuple0.py > tuple_0

```python
tuple_0=("virendra","pradeep",12,True,5.6)
print(type(tuple_0))

# methods of tuple
co=tuple_0.count(5.6)
print(co)
ind=tuple_0.index("pradeep")
print(ind)
# tuples are immutable
# tuple_0[0]="xyz" # this will give error
print(tuple_0)
# slicing of tuple
print(tuple_0[4:])
print(tuple_0[0:3])

# concatenation of tuple
tuple1=(1,2,3)
tuple2=("a","b","c")
tuple3=tuple1+tuple2
print(tuple3)
print(tuple3*2)

# repetition of tuple
tuple4=("virendra",)*5
print(tuple4)
# nested tuple
tuple5=(1,2,(3,4),("a","b"))
print(tuple5)
print(tuple5[2][0]) # accessing nested tuple

# unpacking of tuple
```

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EXPLORER

CHAPTER-4_LISTS_AND_TUPLES

- 1_lists.py
- 1.png
- methods.py
- practiceSet-4.ipynb
- tuple0.py

methods.py > ...

```python
# lists methods
list1=[1,2,3,10,9,5,6,7,8]
print(list1)

list1.append(47) # this will append item in last
list1.sort() # sort numerical lists
list1.reverse()
print(list1)
list1.insert(len(list1), "xyz")
print(list1)
value=list1.pop(-1) # removes last item
print(value)
print(list1)
```